Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/26/2019
PI Name:	Mueller, Holger Ph.D.		
Project Title:	QWEP Analysis and Optimization		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hm@berkeley.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	510-664-4298
Organization Name:	University of California, Berkeley		
PI Address 1:	Physics Department		
PI Address 2:	366 Le Conte Hall # 7300		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-0001	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 8/01/2018 per JPL (Ed., 5/23/18) NOTE: Extended to 06/28/2018 per U. Israelsson/JPL (Ed., 4/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kohel, James Ph.D. (Jet Propulsion Laboratory) Yu, Nan Ph.D. (Jet Propulsion Laboratory)		
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Task Description:	The equivalence of gravity and accelerated motion is the basis of general relativity's description of gravity as a result of curved space and time. At the same time, dark matter and dark energy show that our understanding of physics is incomplete, and that new discoveries are needed to explain the dominant constituents of the universe. Perhaps these are hints that the standard description of gravity must be modified. This description, however, is incompatible with the quantum theories of the electroweak and the strong interactions, in which space and time exist independently of the forces. It is therefore not surprising that unified theories of gravity and quantum mechanics are expected to contain tiny violations of the gravity-acceleration equivalence. Such violations would manifest themselves as variations between the acceleration of free fall for different types of matter. These variations are considered as being among the most promising areas to look for signals arising from a unification of general relativity and quantum mechanics. ESA's (European Space Agency) quantum weak equivalence principle (QWEP) mission seeks such signals with a strongly improved sensitivity of 10-15 or better. By using quantum interference of ultracold atoms, it will use quantum particles where previous tests used bulk matter. The research proposed here will combine theory and experiment to maximize the sensitivity and scientific impact of QWEP, and minimize the risk. Using the standard model extension (SME) as a theoretical framework, we will study which species of atoms, what quantum states, and what type of analysis might lead to the largest and most interesting signals. The choice of species, however, is equally dependent on experimental considerations. Using our existing atom interferometers, we will experimentally compare different technologies for the matter-wave beam splitters that are suitable for the candidate species, and develop strategies to minimize systematic effects such as rotation and gravity gradients. Based on the experience of Jet Propulsion Laboratory (JPL) and the Space Sciences Laboratory (SSL) at University of California (UC) Berkeley, we will match these technologies to the requirement of space operation. Thereby, this work will apply fundamental theory, atomic physics experiment, and principles of astronautics to a comprehensive study optimization and analysis of the QWEP mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We seek to understand two fundamental questions: What is the difference between a quantum and classical test of the weak equivalence principle? What kind of quantum test would provide the most new information regarding Einstein's principle of equivalence? We have made progress on answering these questions, which have been pursued both theoretically and experimentally. We have also started the effort to look for hints of dark energy in atom interferometry experiments. These experiments have started an international effort looking for new models of dark matter and dark energy, and promise to either discover well-motivated theories, or to rule them out once and for all.
	FINAL REPORT SUMMARY--JANUARY 2019 Atom interferometry (AI) has become interesting as a next-gen technology of choice for high-precision tests of fundamental laws of physics such as measurements of fundamental constants as well as tests of gravity and searches for dark energy. This research has broadly explored this technology and its applications, focusing on applications in fundamental physics. The research and its results are described in a number of publications and previous Task Book reports. See the Cumulative Bibliography link at the Bibliography section and previous Task Book reports, as well as the 2018 annual report below. ANNUAL REPORTING MAY 2018 The science accomplished in this funding period was published in the following major publications. In addition, we made progress on several technical fronts, which will be explained afterwards. Measurement of the fine-structure constant as a test of the Standard Model. Richard H. Parker, Chenghui Yu, Weicheng Zhong, Brian Estey, and Holger Müller, Science 360, 191-195 (2018). Measurements of the fine-structure constant a require methods from across subfields and are thus powerful tests of the consistency of theory and experiment in physics. Using the recoil frequency of cesium-133 atoms in a matter-wave interferometer, we recorded the most accurate measurement of the fine-structure constant to date: a = 1/137.035999046(27) at 2.0 × 10-10 accuracy. Comparison with Penning trap measurements of the electron gyromagnetic anomaly ge - 2 via the Standard Model of particle physics is now limited by the uncertainty in ge - 2; a 2.5s tension rejects dark photons as the reason for the unexplained part of the muon's magnetic moment at a 99% confidence level. Attractive force on atoms due to blackbody radiation. Philipp Haslinger, Matt Jaffe, Victoria Xu, Osip Schwartz, Matthias Sonnleitner, Monika Ritsch-Marte, Helmut Ritsch, and Holger Müller, Nature Physics 14, 257–260 (2018) Objects at finite temperature emit thermal radiation with an outward energy-momentum flow, which exerts an outward radiation pressure. At room temperature, a cesium atom scatters on average less than one of these blackbody radiation photons every 10^8 years. Thus, it is generally assumed that any scattering force exerted on atoms by such radiation is negligible. However, atoms also interact coherently with the thermal electromagnetic field. In this work, we measure an attractive force induced by blackbody radiation between a cesium atom and a heated, centimeter-sized cylinder which is orders of magnitude stronger than the outward directed radiation pressure. Using atom interferometry, we find that this force scales with the fourth power of the cylinder`s temperature. The force is in good agreement with that predicted from an ac Stark shift gradient of the atomic ground state in the thermal radiation field. This observed force dominates over both gravity and radiation pressure, and does so for a large temperature range. Testing sub-gravitational forces on atoms from a miniature, in-vacuum source mass. Matt Jaffe, Philipp Haslinger, Victoria Xu, Paul Hamilton, Amol Upadhye, Benjamin Elder, Justin Khoury, and Holger Müller, Nature Physics 13, 938–942 (2017) Traditional gravity measurements use bulk masses to both source and probe gravitational fields. Matter-wave interferometers enable the use of probe masses as small as neutrons, atoms and molecular clusters, but still require fields generated by masses ranging from hundreds of kilograms to the entire Earth. Shrinking the sources would enable versatile configurations, improve positioning accuracy, enable tests for beyond-standard-model ('fifth') forces, and allow observation of non-classical effects of gravity. Here we detect the gravitational force between freely falling caesium

Task Progress:	atoms and an in-vacuum, miniature (centimetre-sized, 0.19 kg) source mass using atom interferometry. Sensitivity down to gravitational strength forces accesses the natural scale for a wide class of cosmologically motivated scalar field models of modified gravity and dark energy. We improve the limits on two such models, chameleons and symmetrons, by over two orders of magnitude. We expect further tests of dark energy theories, and measurements of Newton's gravitational constant and the gravitational Aharonov–Bohm effect. Limits on Lorentz violation in gravity from worldwide superconducting gravimeters. Cheng-Gang Shao, Ya-Fen Chen, Rong Sun, Lu-Shuai Cao, Min-Kang Zhou, Zhong-Kun Hu, Chenghui Yu, and Holger Müller. Phys. Rev. 97, 024019 (2018) We have investigated Lorentz violation through analyzing tides-subtracted gravity data measured by superconducting gravimeters. At the level of precision of superconducting gravimeters, we have brought up and resolved an existing issue of accuracy due to unaccounted local tidal effects in previous solid-Earth tidal model used. Specifically, we have taken local tides into account with a brand new first-principles tidal model with ocean tides included, as well as removed potential bias from local tides by using a worldwide array of 12 superconducting gravimeters. Compared with previous tests with local gravimeters, a more accurate and competitive bound on space-space components of gravitational Lorentz violation has been achieved up to the order of 10-10. Technical progress The science tasks accomplished in this funding period include four types of advances, (i) the development of a new type of atomic beam splitter that achieves high efficiency with limited laser power, called "spin-dependent kicks," (ii) a study of systematic effects arising from the free-fall motion of the atoms in an atomic fountain, (iii) atom interferometry in optical lattices, and (iv) pushing the limits on atom interferometers that use Bragg diffraction and Bloch oscillations. In addition, together with the Bose-Einstein Condensate Cold Atom Laboratory (BECCAL) science definition team on the German and the U.S. side, the Principal Investigator (PI) is working on defining the science objectives of the planned BECCAL mission. Spin-dependent kicks In collaboration with Paul Hamilton (UCLA), we demonstrated this new kind of beam slitter for atom interferometry, which combines the advantages of Bragg diffraction (large momentum transfer and cancellation of Zeeman and AC-Stark shifts) with the ones of Raman transitions (state labeling and ability to address a wide velocity class of atoms). With this, we demonstrated gravimeters, gradiometers, and a "juggling" atom interferometer that's sensitive to oscillatory accelerations. We realized a "juggling" interferometer geometry which is suitable for looking for oscillating forces and accelerations, achieving up to 51 loops. This confirms the extremely high efficiency of this new method. Systematic effects arising from the free-fall motion of the atoms We were also studying the effects of performing atom interferometers in a freely falling frame that are addressed by beams from a stationary laser. At the part-per-billion level of accuracy, the phase will include a term that is proportional to the Doppler effect in the recoil frequency, causing an apparent frame-dependence of the phase. We found that this frame-dependence is cancelled by a counter-term that's proportional to the opening of the interferometer (which is itself proportional to the recoil frequency divided by the atom mass) and the relative velocity of the atom and the source. The two terms together are frame-independent. These effects are relevant in precision measurements of, e.g., the recoil frequency. Lattice interferometry We are continuing to develop atom interferomery with atoms that are held in optical lattice against gravity. If this works, it will be a tool for measuring proximity forces on atoms, and for surpassing the limitations of tall atomic fountains, which (at 10 meters height) allow experiments for no more than 2.8 seconds. We used an optical lattice in an optical cavity to accomplish the hold, based on the hypothesis that the mode-cleaning effected by the cavity would provide better coherence. We have demonstrated a hold time of 1 second. This time is probably limited by laser noise, both amplitude noise as well as frequency noise that gets converted to amplitude noise by the cavity resonance. Better amplitude stabilization and reducing the laser linewidth should help, and will be tried next. Limits of atom interferometry with Bloch oscillations and Bragg diffraction Bragg diffraction needs a detuning in the low GHz region to achieve high two-photon Rabi frequency, but Bloch oscillations prefer detunings of tens of GHz to limit scattering. This tradeoff used to limit us to 10-photon Bragg diffraction. We finished installation of a new laser for driving Bloch oscillations and demonstrated interference with the same contrast as with the previous laser at N=125 Bloch oscillations for various pulse separation times. We now have independent Bragg and Bloch lasers. Unfortunately, the power of the new Bloch laser is no higher than the power we used previously for Bloch. This means that our original strategy to increase the detuning for Bloch won't work, because we can't keep the lattice depth high enough. We will explore upgrading the laser to enable this eventually. We can, however, use the freedom of adjusting the Bloch and Bragg detunings independently to reduce the detuning for Bragg diffraction. With the new laser, we now demonstrated an interferometer with 18-photon Bragg with >10% contrast so far. We are planning to optimize it further. If we can get >30% contrast, we can try adding Bloch oscillations. This would be nearly a factor of 2 improvement in our total phase, and there's nothing stopping us from going further.
Bibliography Type:	Description: (Last Updated: 01/29/2019)
Articles in Peer-reviewed Journals	Jaffe M, Haslinger P, Xu V, Hamilton P, Upadhye A, Elder B, Khoury J, Mueller H. "Testing sub-gravitational forces on atoms from a miniature in-vacuum source mass." Nature Physics. 2017 Oct;13:938–42. (Published online 3 July 2017) https://doi.org/10.1038/nphys4189 , Oct-2017
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